	sound.
2(a)	Centripetal force acting on the satellite is provided by the gravitational force due to the planet. $\frac{GMm}{(R+h)^2} = \frac{mv^2}{R+r}$ $mv^2 = \frac{GMm}{R+r}$ $E_k = \frac{mv^2}{2} = \frac{GMm}{2(R+r)}$

Qn	Solution
2(b)	GPE of satellite = $\dfrac{-GMm}{R+r}$ E_T = Kinetic energy + Gravitational potential energy Total energy $= \dfrac{GMm}{2(R+r)} - \dfrac{GMm}{R+r}$ $= \dfrac{-GMm}{2(R+r)}$
2(c)	$\dfrac{mv^2}{2} = \dfrac{GMm}{2(R+r)}$ $v = \sqrt{\dfrac{GM}{R+r}}$ $v = \sqrt{\dfrac{6.67\times10^{-11}\times7.35x10^{22}}{1.74\times10^{6}+110\times10^{3}}}$ $= 1630 \text{ m s}^{-1}$
2(d)(i)	From part (a), the KE varies with h according to $1/(R+h)$, which means that a lower value of h for satellite P will have a higher KE.
2(d)(ii)	From part (b), the total energy varies with h according to ($-\dfrac{1}{R+h}$), which means that for a smaller value of h for P gives it a more negative total energy compared to S. The total energy is therefore smaller for P compared to S.
3	